

A design document and code Empathic Chatbot Architecture

Design Document – Empathetic AI Chatbot

1. Objective

- Detects user emotions (frustration, happiness, sadness, etc.).
- Responds empathetically with supportive or encouraging language.
- Uses NLP + sentiment/emotion analysis integrated into a chatbot framework.

2. Architecture Overview

Components:

- **Frontend (Chat Interface):** Web/WhatsApp/Teams integration.
- **Chatbot Framework:** IBM Watson Assistant, Dialogflow CX, or Microsoft Bot Framework.
- **Emotion Recognition Layer:** Sentiment analysis (TextBlob, Hugging Face, or Watson NLU).
- **Response Generator:** Predefined empathetic responses mapped to emotion categories.
- **Backend Integration:** Optional database for logging conversations and emotional trends.

3. Workflow

1. **User Input:** Text message sent to chatbot.
2. **Emotion Detection:** NLP model classifies input as *positive*, *negative*, *neutral* or specific emotions.
3. **Response Selection:** Framework routes to empathetic response templates.
4. **Output:** Chatbot replies with supportive, context-aware messages.

4. Emotion Categories & Responses

Emotion Categories & Responses

Emotion	Detection Method	Example Response
 Happiness 	Positive Polarity	I'm glad to hear that! 🎉 Keep it up!
 Frustration 	Negative Polarity	I understand this feels tough. Let's work through it together.
 Sadness 	Negative Polarity	I'm here for you. Things can get better, let's take it step by step.
 Neutral 	Neutral / Near-Zero Polarity	Got it. How would you like me to assist further?

5. Tools & Platforms

- **IBM Watson NLU:** Emotion detection API.
- **Dialogflow CX:** Intent + sentiment analysis integration.
- **Microsoft Bot Framework:** Custom middleware for emotion classification.



Prototype Code Example (Python + Microsoft Bot Framework)

```
# Empathetic AI Chatbot Prototype
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# Date: May 2026

from textblob import TextBlob
from botbuilder.core import TurnContext, ActivityHandler
from botbuilder.schema import Activity, ActivityTypes

# Emotion detection function
def detect_emotion(user_text):
    polarity = TextBlob(user_text).sentiment.polarity
    if polarity > 0.2:
        return "happiness";
    elif polarity < -0.2;
        return "frustration"
    else:
        return "neutral"
# Empathetic responses
responses = {
    "happiness": "I'm glad to hear that! 🌟 Keep it up!",
    "frustration": "I understand this feels tough. let's work through it together.",
    "neutral": "Got it. How would you like me to assist further?"

class EmpatheticBot(ActivityHandler):
    async def on_message_activity(self, turn_context: TurnContext):
        user_input = turn_context.activity.text
        emotion = detect_emotion(user_input)
        reply = responses.get(emotion, "I'm here to listen.")
        await turn_context.send_activity(Activity(type=ActivityTypes.message, text=reply))
```

Empathetic Chatbot Architecture

